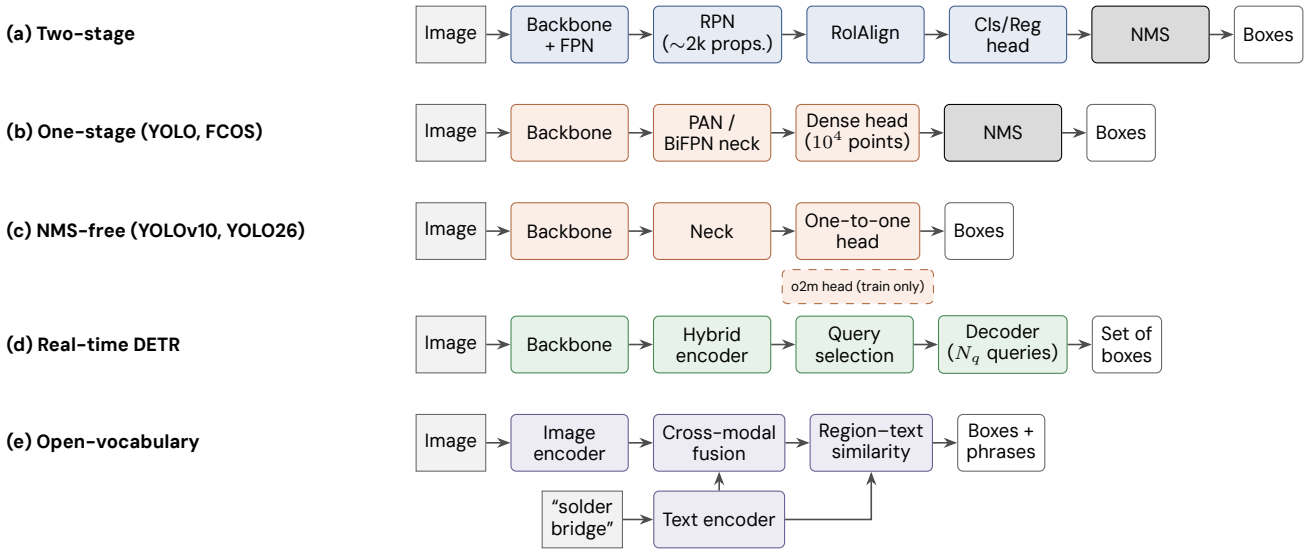
(a) Two-stage
Image
Backbone + FPN
RPN (∼2k props.)
RoIAlign
Cls/Reg head
NMS
Boxes
(b) One-stage (YOLO, FCOS)
Image
Backbone
PAN / BiFPN neck
Dense head (10^4 points)
NMS
Boxes
(c) NMS-free (YOLOv10, YOLO26)
Image
Backbone
Neck
One-to-one head
Boxes
o2m head (train only)
(d) Real-time DETR
Image
Backbone
Hybrid encoder
Query selection
Decoder (N_q queries)
Set of boxes
(e) Open-vocabulary
Image
Image encoder
Cross-modal fusion
Region–text similarity
Boxes + phrases
"solder bridge"
Text encoder